NGBOX-NGLINQ STANDARD CONFIGURATION GUIDE

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CAIM SEA

Doc History

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1. Premise

The NGBox system is a hardware solution that integrates a dedicated computer with specific network components to create a managed and secure connectivity environment. The system consists of two main components:

1. **NGBox:** A specialized computer that serves as the system's central unit. It comes with a pre-configured Windows operating system through Clonezilla and requires standard peripherals such as keyboard, mouse, and monitor (via HDMI) for its initial configuration.
2. **NGLinq:** A USB device that, once connected to the NGBox, provides advanced management and connectivity features through the NG Gateway Admin Tool.

The system is completed by a MikroTik router that manages network connections and is configured with specific settings to ensure proper integration with NGBox and NGLinq.

The combination of these components creates a platform that offers:

- Centralized connectivity management
- Standardized and replicable configuration
- Managed activation and licensing system
- Remote monitoring through TeamViewer
- Network security through dedicated firewall configurations

The system configuration follows a standardized procedure that ensures the proper installation and activation of all components, ensuring that each unit is correctly identified, activated, and configured according to the required specifications.

This solution is particularly suitable in contexts where professional and secure network connectivity management is required, with the possibility of remote administration and constant system monitoring.

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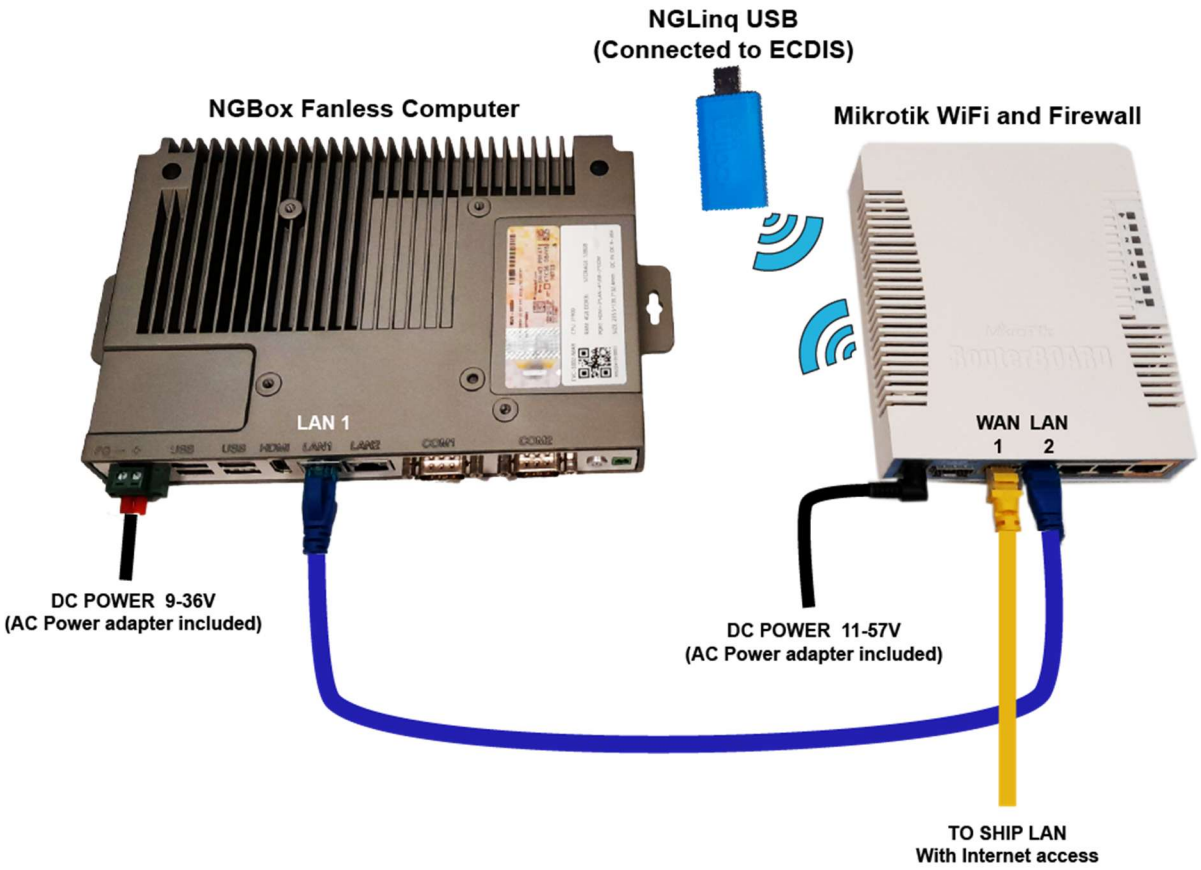
2. Connections



- Place the Mikrotik WiFi Router not too far from the ECDIS, to allow WiFi connection to the NGLinq USB Stick that will be connected to the ECDIS.
- Connect LAN 1 from NGBox computer to Port 2 (LAN) on Mikrotik WiFi Router
- Connect Port 1 (WAN) on Mikrotik WiFi Router to the ship LAN with Internet access
- Provide power (DC POWER 9-36V – AC Power Adapter Included) to the NGBox computer and to the Mikrotik WiFi Router
- Connect the NGLinq USB Stick to the ECDIS and verify that after some seconds the green light inside the stick is on. The green light indicates that the WiFi link with the Mikrotik Router is established.

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3. Credentials

NGBox Computer user: **admin** - pw: **Ngbox_2024!**

Mikrotik Router: login: **admin** - pw: **Ngbox_2024!**

NGBox is provided with TeamViewer for remote assistance. If necessary, ask the credentials to CAIM support.

4. Network Default Configuration

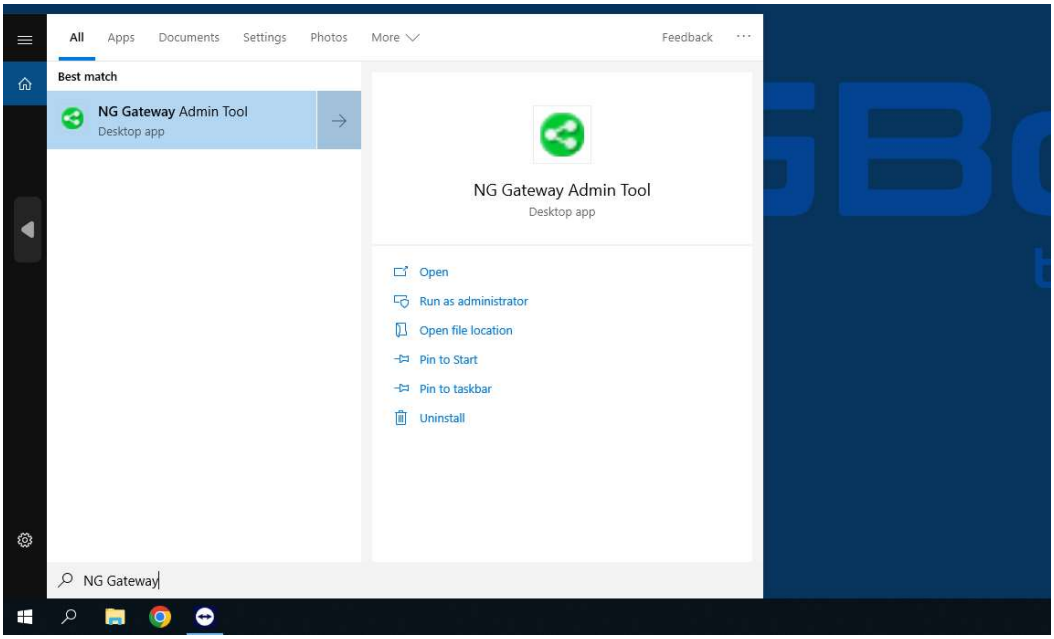
NGBox PC	LAN 1: static IP 172.16.10.20	*LAN2: assigned by DHCP (If the Ship operates in DHCP)*
Mikrotik	LAN Ports 2-5: 172.16.10.1	**The WAN Port 1 is DHCP client (IP assigned by ship)**
NGLinq	WiFi static IP: 172.16.11.30	

5. NGBox Activation

By default, the NGBox software is not activated. You can ask the contract number and activation code for your vessel to CAIM Support team. When the software is not activated, the ENC charts are not downloaded to the NGLinq USB stick.

To activate the NGBox:

1. Access the NGBox PC locally, with monitor and keyboard or remotely using TeamViewer/ RDP
2. Search the **NG Gateway Admin Tool** using Windows search function and run it



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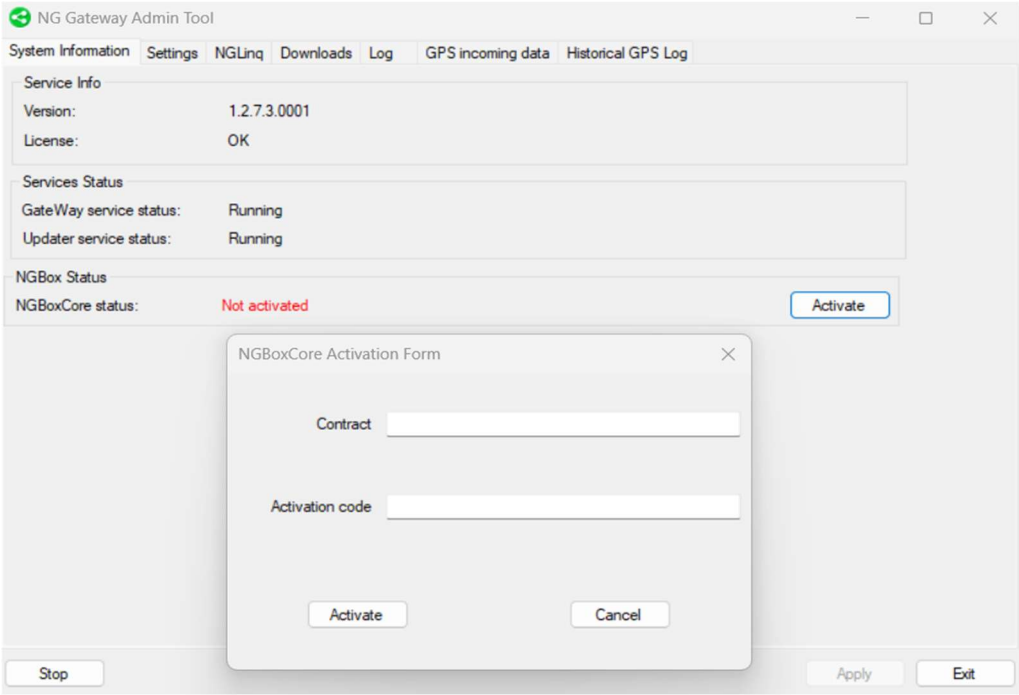
3. In the **System Information** tab, press the **Activate** button, enter the contract number and the activation code and press **Activate.** The activation will fail if there is no Internet access to the server. From now on the version will be 1.2.7.3. This is first thing that needs to be checked.



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6. NGLinq

NGLinq will be pre-activated.

NGLinq shall be plugged close to the MikroTik (we highly suggest 5 meters maximum range).

NGLinq shall turns in fixed green to be sure that WiFi is working properly.

Fixed GREEN	CONNECTED	NGLinq is connected to CAIM WiFi network with stable, reliable connection. System is ready for data exchange operations.	✅ **Normal Operation** No action required
Blinking GREEN	UNSTABLE	**NGLinq is connected to CAIM WiFi but connection is unstable or intermittent. May experience data transfer delays or interruptions.**	⚠️ **Monitor Connection** • Check NGLinq proximity to MikroTik • Verify WiFi signal strength • Check for interference
Blinking BLUE	DATA TRANSFER	NGLinq is actively receiving data via WiFi connection. AVCS charts or updates are being downloaded from NGBox.	📥 **Transfer in Progress** Wait for completion Do not disconnect device
Fixed BLUE	DATA READY	NGLinq has successfully received all data via WiFi. AVCS charts are downloaded and ready for ECDIS installation.	✅ **Ready for Use** Connect to ECDIS for chart installation
Fixed RED	CONNECTION ERROR	**WiFi connection is not available or lost. NGLinq cannot communicate with NGBox or MikroTik router.**	🔧 **Troubleshooting Required** • Check MikroTik router power/status • Verify NGLinq within 5m range • Check WiFi network configuration • Restart MikroTik if necessary
LED OFF	DEVICE OFF	NGLinq device is not powered or not properly connected to USB port. No system activity detected.	🔌 **Check Connection** • Verify USB connection to ECDIS • Check USB port functionality • Try different USB port if available

Quick Reference Notes

- **Optimal Range:** Keep NGLinq within 5 meters of MikroTik router for best performance
- **Power Requirements:** NGLinq is powered through USB connection to ECDIS
- **Data Transfer Time:** Varies based on chart quantity and network connectivity
- **Normal Sequence:** GREEN (connected) → BLUE blinking (downloading) → BLUE fixed (ready)

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7. Prerequisites Need to Done by IT Onboard

7.1 UKHO IP Addresses, Ports and URLs that need to be whitelisted

ADP
IP Address: 137.135.243.254
Port: 80 and 443
Domain Name: www.admiralty.co.uk
URLs:
https://www.admiralty.co.uk/UPDATES/ADP/updating
https://adpactivation.ukho.gov.uk - Utilises 62.172.108.45 Ports 20-21
https://www.admiralty.co.uk/UPDATES/ADP/

ADC
IP Address: 88.198.255.183
Port: 80
Domain Name: www.chartworld.com
URL: http://www.chartworld.com/Files/cb/ac-1-6-0/ac/ADC_Full.zip

FSS File Share Service
URL: https://fss.admiralty.co.uk/

e-NPs
IP Address: 62.172.108.10 and 137.135.243.254
Port: 80 and 443
Domain Name: www.enavigator.ukho.gov.uk (http://www.enavigator.ukho.gov.uk)
URLs: sectigo.com https://enavigator.ukho.gov.uk (https://enavigator.ukho.gov.uk/)
ftp://ukho.gov.uk

Please be aware that IP Addresses can change and therefore users should not restrict their Internet settings to IP addresses but to URLs instead.

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7.2 CAIM Navigate System Requirements for Main and Slave PCs

NaviGate is a software suite for Microsoft Windows consisting of three applications:

1. **NaviGate.exe** the main program with a graphical user interface.

2. **nGateway.exe** the communication gateway with the remote server. It runs continuously as a Windows service.

3. **nGatewayManager.exe** the administration tool to configure gateway settings.

- Minimal Windows version supported is Windows 7 and the software has been tested on up to Windows 11.
- A dedicated PC/Laptop (only CAIM or UKHO softwares are allowed into Navigate PC)
- Minimum processor is an Intel quad core i3 10th generation or higher
- Minimum RAM is 16GB for Windows 10 and above
- Recommended display resolution is 1360 x 768 pixels or higher
- Space required on the hard drive is approx. 250MB for the applications and at least 5GB for the data archive.
- A dedicated WiFi or LAN

NaviGate requires at least .NET Framework 4.8 runtime. The Framework is included in the installer.

We recommend the Navigate PC to be dedicated to this purpose and if possible with dedicated connectivity.

Architecture: standard NaviGate is running on a PC connected to the Internet.

7.3 CAIM Navigate IP Addresses, Ports and URLs that need to be whitelisted

Firewall Setup

In order to exchange data with our server, NaviGate should be able to use the following IPs and ports:

SERVER	PROTOCOL	PORT	IP	APPLICATION
maritimegate.net	HTTPS	443	52.30.174.254	nGateway.exe

Even if not recommended, it is possible to force the program to use HTTP on port 80

SERVER	PROTOCOL	PORT	IP	APPLICATION
maritimegate.net	HTTP	80	52.30.174.254	nGateway.exe

Please contact your IT department and ask to adjust the network/satcom firewall settings and if required, Windows firewall settings.

If the HTTPS protocol is not supported by the vessel, it is possible to switch back to simple HTTP using the NG Gateway Admin Tool and replacing the server address "https://maritimegate.net" with "http://maritimegate.net".

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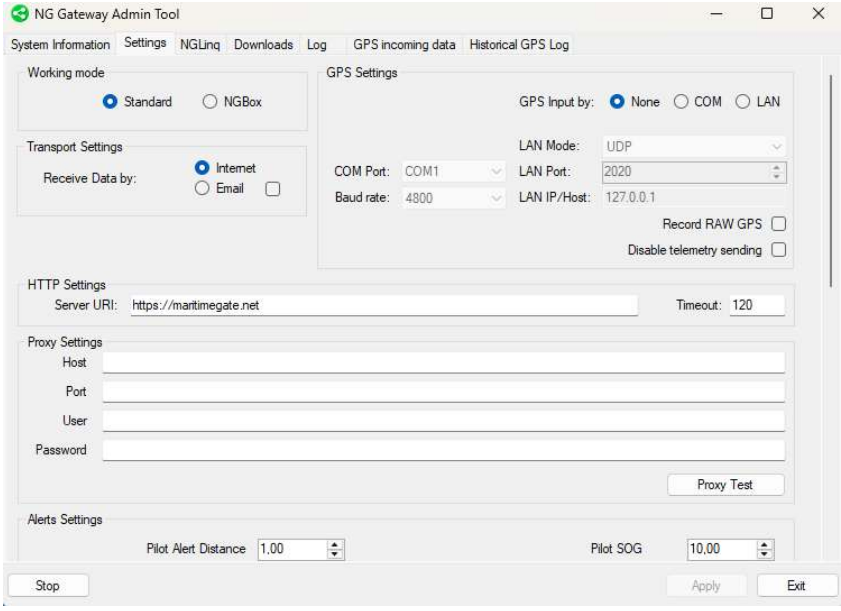

7.3.1 Navigate Software Installation for Main and Slave PCs

To install NaviGate you need Administrator rights. Please contact your IT department to be able to log into your PC as an Administrator. Then run the **NaviGate_Setup.exe** installer.

IMPORTANT: to activate Navigate stations (Main and Backup) contact digital@caim.it to receive activation codes and dedicated instructions.

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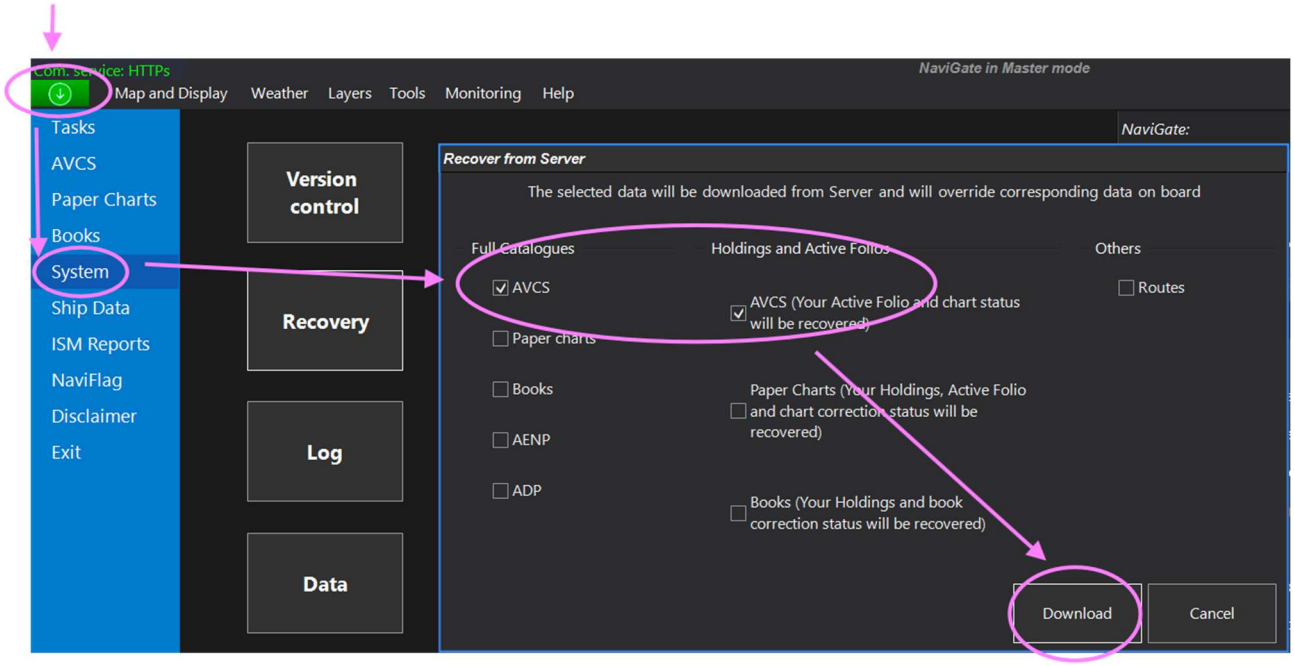
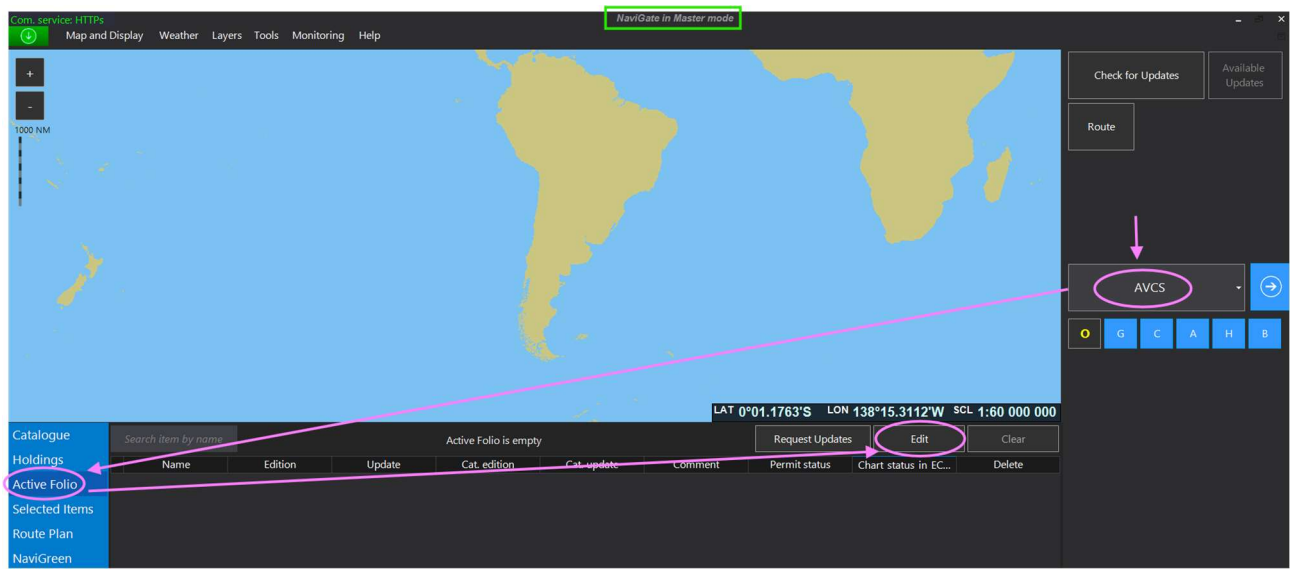
8. First Run Procedure with AVCS PAYS

1. Open Navigate Main Station
2. Do a RECOVERY of AVCS Catalogue and ACTIVE FOLIO



3. EDIT AVCS Active Folio adding charts as per vessel's operative area or per route
4. Once added the needed charts > STOP EDIT > SAVE CHANGES
5. Navigate will exchange your choice with Naviserver ashore and it will receive AVCS permits
6. Action on REQUEST UPDATES button to download the relevant AVCS DATASET



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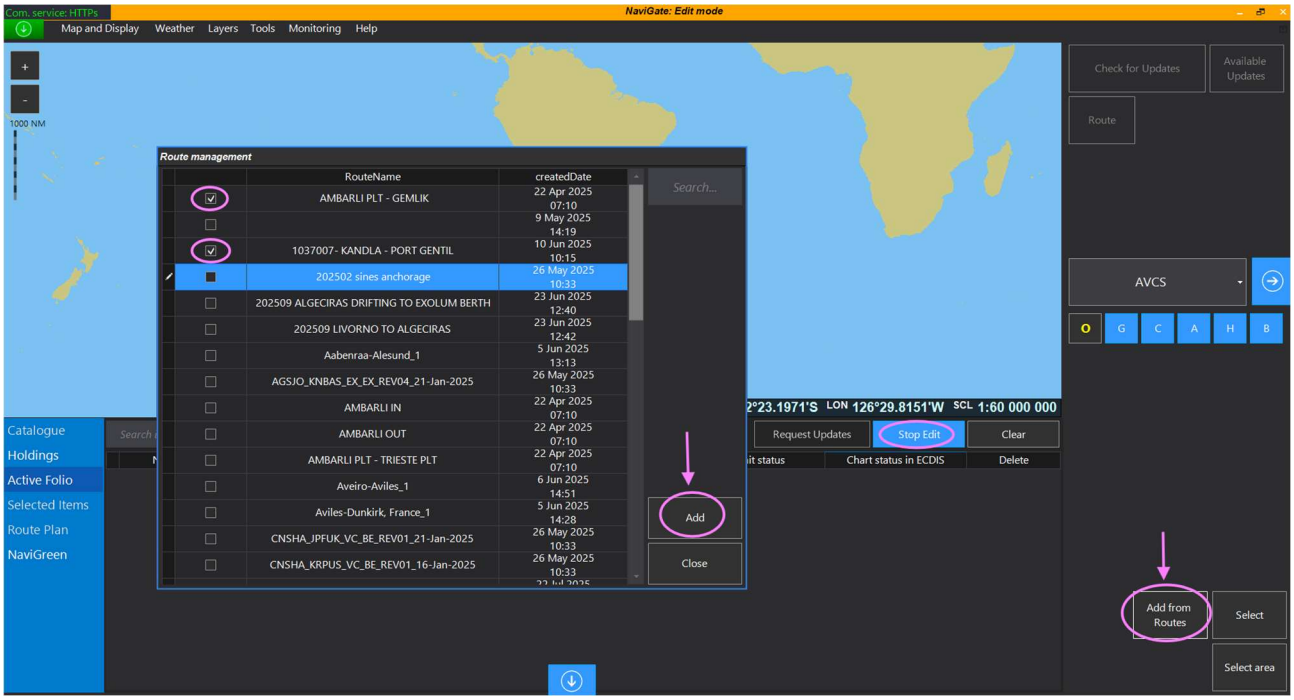

7. Wait for charts to be downloaded, installed and up to date.
8. Do a RECOVERY of AVCS Active Folio again to trigger the NGLinq.
9. NGBOX will download automatically AVCS from Naviserver.

   NOTE: timing will depend by amount of charts and ship connectivity.
10. NGBOX will send to NGLinq the downloaded AVCS, during this phase NGLinq is blinking blue.
11. Once done NGLinq is with a fixed blue led and it is ready for AVCS installation into ECDIS.

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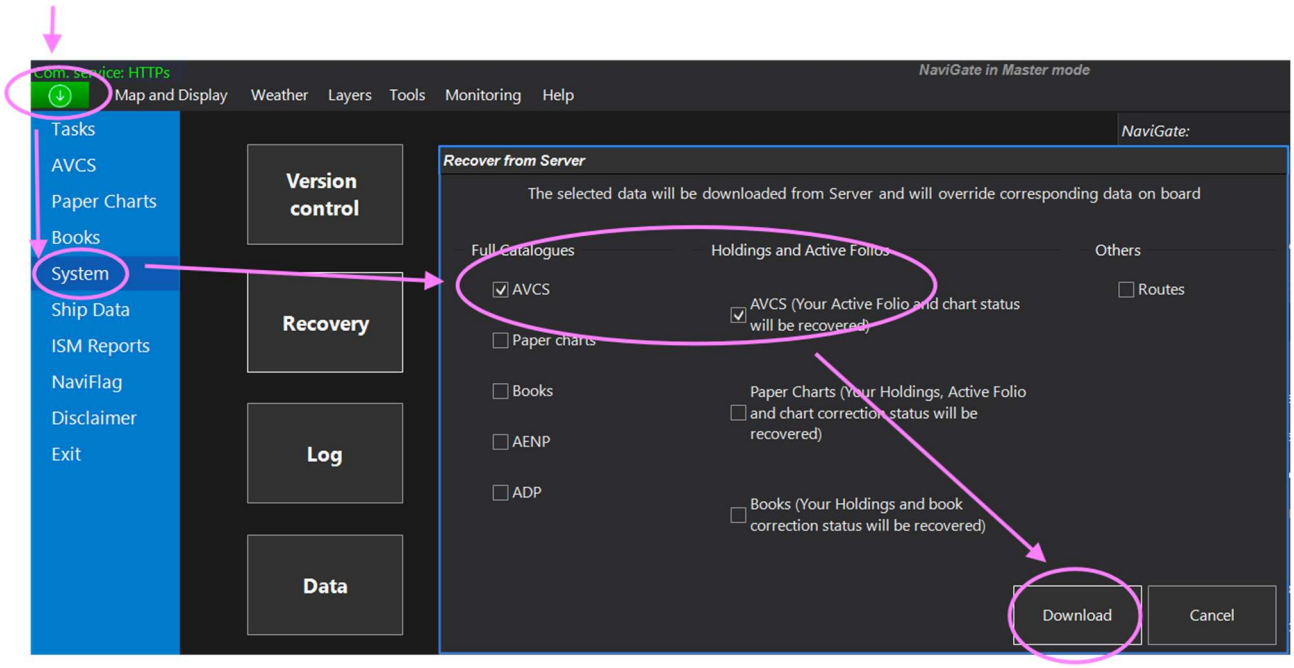
9. First Run Procedure with Standard AVCS

1. Open Navigate Main Station
2. Do a RECOVERY of AVCS Catalogue and ACTIVE FOLIO to import your actual AVCS license

   NOTE: if the vessel had an existing AVCS license, it will be imported as is, if the vessel is brand new, at least 1 AVCS chart shall be purchased (please liaise with digital@caim.it)



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10. Caim Digital Support

Business Hours:

- Monday to Friday: 08:30-17:30 (CET)

- Digital Support Line: +39 010 41 00 299

Emergency Support for Navigation Data (AVCS, ADP, AENP):

1. Send an email to digital@caim.it

2. Then call Our Line +39 010 41 00 299 to advise operators on duty

Digital Support Policy

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